

Lesson 2 ROS2 Installation

Note: The provided Raspberry Pi system image has already set up ROS2 environment. This lesson is offered for your reference only.

This lesson takes the example of humble (requires Internet connection)

1) Press “**Ctrl+Alt+T**” to open a command line terminal and enter the command “**docker pull ros:humble**” to download ROS2 image. The download needs to take a while. Please wait patiently.

```
pi@raspberrypi:~ $ docker pull ros:humble
```

2) After the image is downloaded, enter the command “**docker run -it --network=host -d -v=/dev:/dev -v /tmp/.X11-unix:/tmp/.X11-unix --name humble -e DISPLAY=\${DISPLAY} --restart=always ros:humble /bin/bash**” in the terminal to run the container and specify the name as “humble”.

```
pi@raspberrypi:~ $ docker run -it --network=host -d -v=/dev:/dev -v /tmp/.X11-unix:/tmp/.X11-unix --name melodic -e DISPLAY=${DISPLAY} --restart=always ros:melodic /bin/bash
75b6634e622762255058569dd769ed3925f77b9976539e85d1d70cd029929f9d
pi@raspberrypi:~ $ docker ps
```

3) Enter “**xhost +**” to start the access control for X Sever.

```
pi@raspberrypi:~ $ xhost +
access control disabled, clients can connect from any host
```

4) Enter “**docker ps**” to view the ID of the newly created container.

```
pi@raspberrypi:~ $ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
55ce8db5f027	ros:humble	"/ros_entrypoint.sh ..."	2 days ago	Up 21 hours		humble
1318ca2bfbd5	ros:melodic	"/ros_entrypoint.sh ..."	2 days ago	Up 21 hours		melodic

```
pi@raspberrypi:~ $
```

5) Enter the command “**docker exec -it 55ce /bin/bash**” to enter the container (the container’s ID can be abbreviated as long as it uniquely

identifies the container).

```
pi@raspberrypi:~ $ docker exec -it 55ce /bin/bash
```

6) Enter the command “**useradd -m -s /bin/bash ubuntu**” to create a new user.

```
root@raspberrypi:/# useradd -m -s /bin/bash ubuntu
```

7) In the terminal, enter “**passwd ubuntu**” to set the password for “ubuntu”. Here, “ubuntu” is set as the password, you will be prompted to re-enter the password.

```
root@50e095a38fff:/# passwd ubuntu
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
```

8) Enter “**usermod -aG sudo ubuntu**” to add the new user “ubuntu” to the sudo group, granting it the superuser permission.

```
root@raspberrypi:/# usermod -aG sudo ubuntu
```

9) Enter “**sudo apt-get update -y && sudo apt-get upgrade -y**” to update the list of available software packages and upgrade the installed software packages on the system.

```
root@raspberrypi:/# sudo apt-get update -y && sudo apt-get upgrade -y
Get:1 http://snapshots.ros.org/melodic/final/ubuntu bionic InRelease [13.8 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports bionic InRelease [242 kB]
Get:3 http://snapshots.ros.org/melodic/final/ubuntu bionic/main arm64 Packages [1017 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports bionic-updates InRelease [88.7 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports bionic-backports InRelease [83.3 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports bionic-security InRelease [88.7 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports bionic/universe arm64 Packages [11.0 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports bionic/main arm64 Packages [1285 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports bionic/restricted arm64 Packages [572 B]
```

10) Enter “**sudo apt-get install vim -y**” to install Vim text editor.

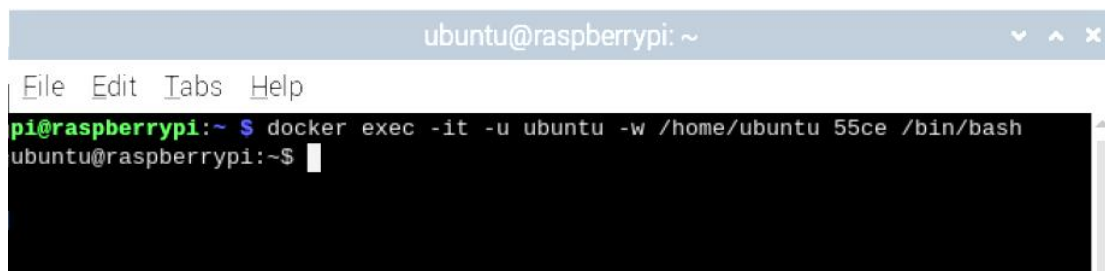
```
root@raspberrypi:/# sudo apt-get install vim -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libgpm2 vim-common vim-runtime xxd
Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
```

11) Enter “**sudo apt-get install ros-humble-desktop-full -y**” to install the complete desktop environment of ROS Humble version.

```
root@raspberrypi:/# sudo apt-get install ros-melodic-desktop-full -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  autoconf automake autopoint autotools-dev avahi-daemon bind9-host blt bsdmainutils curl cython dbus dh-autoreconf dh-strip-nondeterminism dmsetup file fltk1.3-doc fluid fonts-lato fonts-liberation font
```

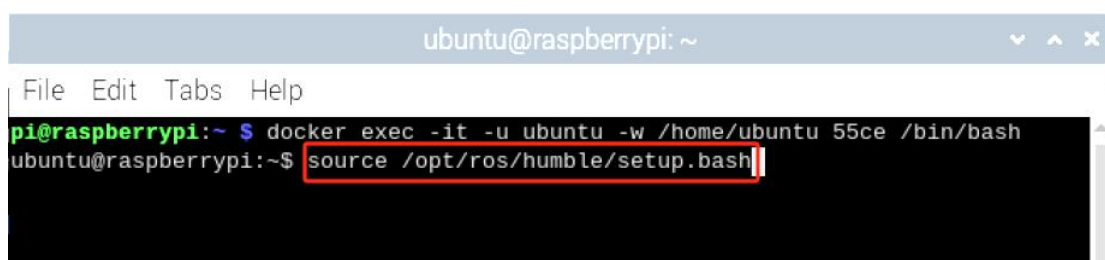
2. Test ROS2 Environment

1) Enter “**docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash**” to enter the container. (Note: 55ce is the container ID with the ROS2 environment installed.)



The screenshot shows a terminal window titled "ubuntu@raspberrypi: ~". The user is at the prompt "pi@raspberrypi:~" and has entered the command "docker exec -it -u ubuntu -w /home/ubuntu 55ce /bin/bash". The output shows the user is now inside the container at the prompt "ubuntu@raspberrypi:~\$".

2) Enter “**source /opt/ros/humble/setup.bash**” to manually set up ROS2 environment.



The screenshot shows the same terminal window as before, but now the user is at the prompt "ubuntu@raspberrypi:~\$" and has entered the command "source /opt/ros/humble/setup.bash". The command is highlighted with a red box.

3) The step 2 needs to be executed every time to load the workspace when you open the terminal, you can enter “**echo “source /opt/ros/humble/setup.bash” >> ~/.bashrc**” to write this command into the .bashrc file.

```
ubuntu@raspberrypi:~$ echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
```

4) Enter “**source ~/.bashrc**” to make the .bashrc file take effect. In this way, there is no need to load the workspace environment for each operation.

```
ubuntu@raspberrypi:~$ source ~/.bashrc
ubuntu@raspberrypi:~$
```

5) Enter “**ros2 run turtlesim turtlesim_node**” to start the TurtleSim GUI interface. If it launches successfully, it indicates that ROS2 has been installed successfully.

```
ubuntu@raspberrypi:~$ ros2 run turtlesim turtlesim_node
```

